

1. Administrative & Study Identification

Full Study Title: A Randomized, Open-label, Multi-center Phase III Trial Comparing Tisagenlecleucel to Standard of Care in Adult Participants With Relapsed or Refractory Follicular Lymphoma
Official Regulatory Title: A Phase III Trial Comparing Tisagenlecleucel to Standard of Care (SoC) in Adult Participants With r/r Follicular Lymphoma
Short Title/Acronym: LEDA Study
Protocol Number: CCTL019E2301 (Secondary ID: 2023-503452-27-00)
NCT Number: NCT05888493 **Posting ID:** 422866721853332151 **Sponsor / Company Name:** Novartis Pharmaceuticals **Investigational Product:** Tisagenlecleucel (tisa-cel / CD19-directed CAR-T cell therapy)
Phase of Development: Phase III **Trial Status:** Recruiting **Medical Monitor:** Novartis Clinical Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective
Primary Objective: To demonstrate the superiority of the tisagenlecleucel treatment strategy over standard of care (SOC) therapy with respect to progression-free survival (PFS) determined by blinded independent review committee (BIRC) based on the Lugano response criteria.
Secondary Objectives: To evaluate the complete response rate (CRR), overall response rate (ORR), overall survival (OS), duration of response, and overall safety/tolerability profile.

2.2 Background & Rationale Follicular lymphoma (FL) is an indolent disease considered incurable with standard chemoimmunotherapy (CIT). Patients with relapsed or refractory (r/r) FL after two or more lines of systemic therapy face a poor prognosis and limited durable treatment options. Tisagenlecleucel has previously demonstrated significant clinical benefit in B-cell malignancies. This study aims to establish definitive evidence of tisagenlecleucel's superiority over standard therapies (R2 or R-CHOP) to improve progression-free survival and establish a new standard of care in the third-line or later setting.

3. Study Design & Methodology

3.1 Design Framework
Study Type: Interventional **Control Type:** Active Comparator (Standard of Care: R2 or R-CHOP based on investigator choice prior to randomization)
Blinding: Open-label (with Blinded Independent Review Committee [BIRC] evaluating the primary endpoint)
Randomization: Randomized, Parallel Group

3.2 Planned Enrollment & Duration Target ≥ 108 adult participants **Number of Sites:** Global, Multi-center **Estimated Study Start:** January 2024 **Estimated Primary Completion:** Q1 2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years at the date of signing the informed consent form. 2. Follicular lymphoma grade 1, 2, or 3A confirmed histologically after the latest relapse (local assessment). 3. Relapsed or refractory disease after a second or later line of systemic therapy including an anti-CD20 antibody and an alkylating agent. 4. Disease that is both active on Positron emission tomography (PET) scan (defined as a score of 4 or 5 on the Deauville 5-point scale) and measurable on Computed tomography (CT) scan. 5. ECOG performance status of 0, 1, or 2 at screening. 6. Adequate hematologic, renal, hepatic and pulmonary organ function at screening. 7. Must meet the institutional criteria to undergo leukapheresis (unless historical leukapheresis is available). 8. Must be eligible for treatment with the selected standard of care regimen.

4.2 Exclusion Criteria 1. Follicular lymphoma grade 3B or evidence of histologic transformation. 2. Prior treatment with anti-CD19 therapy, gene therapy, or adoptive T-cell therapy. 3. Active CNS involvement by malignancy. 4. Clinically significant active infection, presence of Human immunodeficiency virus (HIV) antibody, or active hepatitis B or C. 5. Active neurological autoimmune or inflammatory disorders (e.g., Guillain-Barré syndrome). 6. Investigational medicinal product within the last 30 days or five half-lives (whichever is longer) prior to randomization. 7. Clinically significant cardiovascular conditions such as acute coronary syndrome, significant cardiac arrhythmias, heart failure or decreased LVEF.

5. Intervention & Treatment Plan

5.1 Investigational Regimen (Arm A) Dosage/Frequency: Single infusion of 0.6 to 6×10^8 CAR-positive viable T-cells. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Single infusion following required leukapheresis and lymphodepletion.

5.2 Concomitant & Prohibited Therapy (Arm B / SOC) Permitted: Active comparator standard of care (Lenalidomide + rituximab [R2] for up to 12 cycles, or Rituximab + CHOP [cyclophosphamide, doxorubicin, vincristine, and prednisone] for 6 to 8 cycles) for Arm B based on investigator choice determined prior to randomization. Standard supportive therapy for CAR-T toxicity (e.g., tocilizumab/corticosteroids for cytokine release syndrome). **Prohibited:** Use of any investigational medicinal

product within the last 30 days or five half-lives (whichever is longer) prior to randomization.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Eligibility Assessment, PET/CT scans, Leukapheresis qualification
Baseline	Day 0	Randomization, Leukapheresis collection (Arm A) or Start of SoC (Arm B)
Interim	Varies	Tisa-cel Infusion, close monitoring for CRS/ICANS, Efficacy response imaging
End of Study	PD / Long-term	Documentation of disease progression, Survival follow-up, Long-term safety tracking

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS), defined as the time from randomization to objective disease progression (by BIRC) or death from any cause. **Measurement Method:** Evaluated by a Blinded Independent Review Committee (BIRC) based on the Lugano response criteria.

7.2 Secondary & Exploratory Endpoints 1. Complete Response Rate (CRR) as assessed by BIRC (Key Secondary). 2. Overall Response Rate (ORR) by BIRC. 3. Overall Survival (OS). 4. Incidence, nature, and severity of treatment-emergent adverse events (e.g., CRS, ICANS).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Intensive inpatient/outpatient monitoring post-infusion specifically targeting Cytokine Release Syndrome (CRS) and Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS), as well as prolonged cytopenias. **Serious Adverse Events (SAEs):** Expected 24-hour reporting threshold for life-threatening events per standard oncology protocols. **Data Monitoring Committee (DMC):** An Independent Data Monitoring Committee (iDMC) will evaluate safety and efficacy data systematically.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy evaluations; Safety Analysis Set for participants receiving any study treatment. **Sample Size Justification:** Study powered ($\alpha = 0.05$, expected Power $\geq 80\%$) to detect a statistically significant PFS advantage of

Tisagenlecleucel over SOC therapies. **Handling of Missing Data:** Censoring applied for PFS per standard FDA/EMA oncology data guidelines (censored at last evaluable disease assessment prior to initiation of new anti-cancer therapy or missing visits).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Rigorous onsite and remote monitoring with particular attention to leukapheresis chain-of-identity, product handling, and AE reporting. **Audit Trail:** Validated eCRF environment with robust data clarification processes and centralized radiological image storage for BIRC review.

11. Study Identifiers & Governance

Primary ID: CCTL019E2301 **Registry Number:** NCT05888493 **Sponsor/Lead Organization:** Novartis **Study Title:** LEDA **Disease Areas:** Follicular Lymphoma (FL)

12. Clinical Framework (Oncology/Lymphoma Specific)

Primary Condition: Relapsed or Refractory Follicular Lymphoma (r/r FL) **Diagnosis Threshold:** Histologically confirmed grade 1, 2, or 3A with Deauville Score 4 or 5 **Medical Methodology:** Cellular Immunotherapy (CAR-T) vs. standard Chemoimmunotherapy (CIT) **Optimization Target:** Eradication of detectable active disease and prolongation of PFS

13. Study Procedures & Methodology

Management Pathway: Step-wise pathway involving apheresis, bridging therapy (if needed), lymphodepletion, CAR-T infusion, and strict toxicity management (Arm A) compared to cyclic standard drug administration (Arm B). **Education Protocol (HCP):** REMS (Risk Evaluation and Mitigation Strategy) training, specific guidelines for CRS/ICANS identification and treatment (e.g., grading by ASTCT criteria). **Education Protocol (Patient):** Comprehensive pre-infusion counseling regarding expected immediate toxicities and required proximity to certified treatment centers post-infusion. **Quality Audit Mechanism:** Independent central review of PET/CT images to ensure objective response evaluation.

14. Observation & Follow-Up Schedule

Total Duration: Standard initial efficacy phase up to ~2-3 years; Long-term safety follow-up required up to 15 years per regulatory guidelines for gene therapies. **Onsite Visit Cadence:** Highly frequent during the first 28 days post-infusion; transition to Month 3, Month 6, and quarterly thereafter. **Remote Monitoring Frequency:** Intermittent telehealth and lab draw checkpoints between physical visits during the maintenance phase. **Checklist Review Interval:** Routine survival sweeps every 12 weeks (Q12W) post-disease progression.

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator		Dr. Shafqat Inam / [Site PI]		_____		[DD-MMM-YYYY]			Clinical Operations Lead		[Name]		_____		[DD-MMM-YYYY]	
			Lead Statistician		[Name]		_____										